

Principles of Economics

Problem Set 3

Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP QUESTIONS

1. Identify a social dilemma that related to climate change and specify how it satisfies the definition of a social dilemma.
2. What is the likely result of the game in Figure 4.22 (the payoffs are in thousands of \$) if:
 - a. Astrid can choose which language she will use first, and commit to it (just as the Proposer in the ultimatum game commits to an offer, before the Responder responds)?
 - b. The two can make an agreement, including which language they use, and how much cash can be transferred from one to the other?
 - c. They have been working together for many years, and in the past they used Java on joint projects?

		Bettina	
		Java	C++
Astrid	Java	Both work in the same language Astrid benefits more: she is better at Java coding	Each is working in the language they are better at But working in different languages is less productive than if both work in the same language
	C++	Each is working in the language they are less good at, and so neither works fast Working in different languages is less productive	Both work in the same language Bettina benefits more: she is better at C++ coding

		Bettina	
		Java	C++
Astrid	Java	3 4	2 2
	C++	0 0	6 3

3. Return to the prisoners' dilemma pest control game that Anil and Bala played in Figure 4.4b, but now suppose that the game is played sequentially, like the ultimatum game. One player

(chosen randomly) chooses a strategy first (the first mover), and then the second moves (the second mover).

- Suppose you are the first mover and you know that the second mover has strong reciprocal preferences, meaning the second mover will act kindly towards someone who upholds social norms not to pollute and will act unkindly to someone who violates the norm. What would you do?
- Suppose the reciprocal person is now the first mover interacting with the person she knows to be entirely self-interested. What do you think would be the outcome of the game?

		Bala	
		IPC	Terminator
Anil	IPC	3, 3	4, 1
	Terminator	1, 4	2, 2

2. MULTIPLE-CHOICE QUESTIONS

4. Select the correct answer:

- The outcome (I, T) is attained as the dominant strategy equilibrium if Anil is completely selfish and Bala is somewhat altruistic.
- The outcome (I, T) is attained as the dominant strategy equilibrium if Anil is somewhat altruistic and Bala is very altruistic.
- The outcome (I, T) cannot be attained as the dominant strategy equilibrium.
- The outcome (I, I) is attained as the dominant strategy equilibrium only if both Anil and Bala are somewhat altruistic.

5. Bruce owns a cooperative project with two other members. Any member that chooses to put in a full day of work faces a cost of £50 but produces a total income of £120, which is shared amongst the three. So, for example, if Bruce and one other member do a full day of work, then the income per member is $(£120 \times 2)/3 = £80$, leaving Bruce with a net income of $£80 - £50 = £30$. Assume that a member must either put in a full day of work or none at all. Based on this information, we can conclude that:

- The socially optimal outcome (one with the highest total net income) is when no one works.
- The dominant strategy equilibrium of this public goods game is when everyone works.
- Bruce is better off not working, irrespective of the actions of the other members.
- Bruce's net income when all three members work is £80.

6. The following game represents the interaction between two software engineers, Astrid and Bettina, who are working together to write code as a part of a project. Astrid is better at writing Java code, while Bettina prefers C++. The numbers represent the pay in thousands of dollars for completion of the project. Based on this information, which of the following are true?

		Bettina	
		Java	C++
Astrid	Java	3 4	2 2
	C++	0 0	6 3

- There are two Nash equilibria: (Java, Java) and (C++, C++).
- If Astrid can choose the format first and commit to it, then (Java, C++) will be chosen.
- If the two can make an agreement beforehand, including a transfer of \$500 from Bettina to Astrid, then (C++, C++) will be chosen.
- If the two can make an agreement beforehand, including a transfer of \$3,500 from Bettina to Astrid, then (C++, C++) will be chosen.

3. CONCEPTUAL QUESTIONS

- Many people consider political advertising (campaign advertisements) to be a classic example of a prisoners' dilemma. Using examples from a recent political campaign with which you are familiar, explain whether this is the case.
- Imagine you are the mayor of a small town and wish to motivate your citizens to get involved in 'City Beautiful Day', in which people spend one day to help cleaning parks and roads. How would you design the day to motivate citizens to take part?
- In class we saw that altruistic preferences can ease a social dilemma: if Anne values Rachel's payoff enough, the equilibrium of the ultimatum game moves from 9–1 to 5–5. It is tempting to conclude that the more people care about each other, the better the outcome will be. The following example shows that the conclusion is too quick. Use it to discuss why.

Beto makes a piece of furniture to order for Ana. He chooses his effort first: high effort costs him 4 and the piece is worth 10 to Ana, while low effort costs him nothing and the piece is worth 3. Ana then sees the quality and decides whether to accept it and pay the agreed price of 6, or to reject it and pay nothing, in which case both of them get 0. Ana is altruistic, so she evaluates the outcome as $u_A = m_A + \alpha m_B$.

What happens as Ana becomes more altruistic, and what does it tell us about the claim above?